ICT 1104 — Information Systems

Complete Study & Revision Guide

Compiled from UCU Test 1, Test 2, Test 3 & Advent 2024 / August 2025 Exams

This guide simplifies every topic that has appeared in your Information Systems tests and exams, then shows you the different ways each topic can be examined — with model answers — so you walk into the exam room recognising the question no matter how it's phrased.

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1. Foundations — What is an Information System?

1.1 Definition

An **Information System (IS)** is a set of interrelated components (people, hardware, software, data, and procedures) that work together to **collect, process, store, and distribute information** to support decision-making, coordination, and control in an organisation.

Simple way to remember it: **People + Technology working together to turn raw data into useful information.**

1.2 Business Strategy Link

Before any information system is built, managers must first understand the organisation's **business strategy** (the plan for how the company will compete and win in its industry) and its **organisational strategy** (how the company structures itself to carry out that plan). The information system is then designed to support that strategy — not the other way round.

Q: Define an information system.

A: An information system is a combination of people, hardware, software, data, and procedures organised to collect, process, store, and distribute information that supports decision-making and control within an organisation.

Q: What is a business strategy?

A: A business strategy is a company's plan of action for how it will compete in its industry, attract customers, and achieve its goals better than its rivals.

Exam Angle — how this can be asked:

- "Define an information system" (short definition, 2 marks)
- "What is a business strategy?" (2 marks)
- "Explain why managers must understand business strategy before systems are developed" (essay style)
- Case-study intro paragraphs (like the "Jupiter Chemical" or "GHS" cases) almost always expect you to apply this definition to the scenario given.

2. DIKW Hierarchy & Knowledge Types

2.1 The DIKW Hierarchy (Data → Information → Knowledge → Wisdom)

This shows how raw facts get turned into wise decisions, step by step:

Level	Meaning	Example
Data	Raw, unorganised facts with no context	"37"
Information	Data that has been processed/organised to have meaning	"Patient's temperature is 37°C"
Knowledge	Information combined with experience/understanding, used to make sense of a situation	"37°C is a normal body temperature"
Wisdom	Applying knowledge with judgement to make the best decision	"The patient does not need fever medication"

> Illustration to draw: four stacked/rising blocks labelled Data → Information → Knowledge → Wisdom, each one built on top of the one before it (like a pyramid), because each level depends on the one below.

2.2 Tacit vs Explicit Knowledge

Tacit Knowledge	Explicit Knowledge
Personal, experience-based knowledge that is hard to write down or transfer (e.g. a doctor's "gut feeling" during diagnosis)	Knowledge that is formally documented and easy to share (e.g. a procedures manual, a textbook)
Learned through practice, observation, mentorship	Learned by reading, training materials, databases
Difficult to store in an information system directly	Easy to store, search, and retrieve in an information system

Q: With the aid of an illustration, explain the DIKW hierarchy.

A: DIKW stands for Data, Information, Knowledge, Wisdom. Data is raw unorganised facts (e.g. numbers, words). Information is data processed and given context so it becomes meaningful. Knowledge is information combined with experience that helps someone understand a situation. Wisdom is applying that knowledge with good judgement to make the right decision. It is drawn as a rising pyramid because each level builds on the one below it.

Q: Differentiate tacit and explicit knowledge.

A: Tacit knowledge is personal, experience-based, and hard to write down or share (e.g. a skilled nurse's intuition). Explicit knowledge is formally documented, structured, and easy to store, search, and share (e.g. hospital procedure manuals stored in a database).

Exam Angle:

- "With the aid of an illustration, explain the DIKW hierarchy" (10 marks — draw the pyramid!)
- "Differentiate tacit and explicit knowledge" (2 marks)
- Could also appear as a matching/scenario question: "Identify which stage of DIKW is represented by X statement in the case"

3. Components & Core Functions of an Information System

3.1 The Five Components of an Information System

Component	What it means	Example
Data	Raw facts fed into the system	Patient names, admission dates
Hardware	Physical devices used to input, process, output, and store data	Computers, scanners, printers, servers
Software	Programs/instructions that tell hardware what to do	Hospital Management System app, Windows OS
People	Users and IT staff who interact with or run the system	Doctors, receptionists, database administrators
Process (Procedures)	The rules and steps for how the system is used	Steps to register a new patient

How they interact: **People** follow **processes** to feed **data** into the system using **hardware**, which is controlled by **software** that processes the data into useful information for decision-making. Remove any one component and the system breaks down.

3.2 The Four Core Functions of an Information System

1. **Input** – capturing/collecting raw data (e.g. typing patient details into a form)
2. **Processing** – converting raw data into meaningful information (e.g. calculating a patient's bill)
3. **Output** – presenting processed information to the user (e.g. printing a report or invoice)
4. **Storage** – saving data/information for future use (e.g. saving patient records in a database)

*Some versions add a 5th function: **Feedback/Control** — monitoring output and correcting input/processing if something is wrong. If a question asks for only 3 core functions, use Input → Processing → Output.*

3.3 Benefits of an Information System (e.g. to a University like UCU)

- **Faster processing** — student registration and results are processed quickly instead of manually
- **Improved accuracy** — reduces human error in records like tuition balances and grades
- **Better decision-making** — management gets real-time reports (e.g. enrolment trends) to plan ahead
- **Improved communication** — email, portals, and notices reach students and staff instantly
- **Cost & time savings** — less paperwork, less duplicated effort
- **Enhanced security** — controlled access to sensitive records like grades and finances

Q: With relevant examples, explain how the components of an information system interact to form an effective information system.

A: The five components — people, hardware, software, data, and process — work together as a chain. For example, in a hospital system: a receptionist (people) follows the patient-registration procedure (process) to enter a patient's details (data) into a computer (hardware) running the hospital software (software). The software processes this data into useful information, such as an appointment schedule, which is then stored for future retrieval. If any component is missing or fails — e.g. no trained staff, or faulty hardware — the whole system cannot function effectively.

Q: With the aid of an illustration, describe the four major functions of an information system.

A: Draw a simple flow: Input → Processing → Output → Storage (with an optional Feedback loop back to Input). Input captures raw data; Processing converts it into meaningful information; Output presents that information to users; Storage keeps data/information for later use; Feedback checks the output and sends corrections back into the system.

Exam Angle:

- "Write short notes on hardware/software/data/feedback/digital world" (2 marks each — see Section 13 for these)
- "Explain any four benefits of an information system to an institution like UCU" (8 marks)
- "With the aid of an illustration, explain the 4/3 core functions of an information system" (4–6 marks — always draw a labelled diagram)
- Diagrams are often worth marks on their own — never skip drawing them even in a rush.

4. Business Strategy & Competitive Advantage

4.1 Competitive Advantage

> **Competitive advantage** is an edge a company has over its rivals that allows it to attract more customers, earn higher profits, or operate more efficiently than the competition — e.g. lower prices, better quality, or faster service.

4.2 Michael Porter's Three Generic Strategies for Competitive Advantage

Strategy	Explanation
Cost Leadership	Becoming the lowest-cost producer in the industry so you can sell at lower prices than competitors while still making a profit (e.g. Walmart, using IS to cut supply-chain costs).
Differentiation	Offering products/services that are noticeably different or better than competitors (unique features, better quality, stronger brand) so customers choose you even at a higher price.
Focus (Niche)	Concentrating on a specific market segment, region, or customer group and serving it better than broad competitors can (e.g. a system tailored only for university students).

4.3 Five Ways Information Systems Help Organisations Gain Competitive Advantage

1. **Lowering costs** — automating tasks reduces labour and operational costs (supports cost leadership)
2. **Improving customer service** — CRM systems personalise service and increase customer loyalty
3. **Differentiating products/services** — e.g. offering online tracking or mobile apps competitors lack
4. **Creating switching costs** — making it inconvenient for customers to move to a competitor (e.g. loyalty data locked into your platform)
5. **Building barriers to entry** — heavy IT infrastructure investment makes it hard for new competitors to enter
6. **Improving speed and efficiency** — faster decision-making using real-time data and analytics

(Pick any 4–5 of the above depending on the marks allocated.)

4.4 Michael Porter's Five Forces Model

This model analyses the competitive pressures in an industry:

1. **Threat of new entrants** — how easily new competitors can enter the market
2. **Bargaining power of suppliers** — how much suppliers can raise prices or reduce quality
3. **Bargaining power of buyers/customers** — how much customers can demand lower prices or better quality
4. **Threat of substitute products/services** — alternative products that satisfy the same need
5. **Rivalry among existing competitors** — intensity of competition among current firms in the industry

> Illustration: draw the target company in the centre with five arrows pointing inward, one labelled for each force above.

Q: Explain the three strategies for achieving competitive advantage according to Michael Porter.

A: (1) Cost leadership — becoming the cheapest producer to undercut rivals on price. (2) Differentiation — making your product/service stand out through quality, features, or branding. (3) Focus — targeting a narrow market segment and serving it better than generalist competitors.

Q: Discuss four ways a company can leverage information systems to achieve a competitive advantage.

A: 1) Cost reduction through automation. 2) Better customer service through CRM/personalisation. 3) Product/service differentiation through added digital features. 4) Creating switching costs so customers stay loyal. (See section 4.3 for more options.)

Exam Angle:

- "What is a business strategy?" / "Describe competitive advantage" (2 marks each)
- "Explain the three strategies for achieving competitive advantage according to Michael Porter" (6 marks)
- "Explain any five different ways information systems support organisations to achieve competitive advantage" (10 marks)
- "With the aid of an illustration, describe Michael Porter's five forces model" (10 marks — always draw the diagram)

5. Types of Information Systems & Quality Information

5.1 Types of Information Systems by Organisational Level

System	Used by	Purpose
TPS – Transaction Processing System	Operational staff	Records day-to-day routine transactions (e.g. sales, payroll, admissions)
MIS – Management Information System	Middle managers	Summarises TPS data into routine reports for monitoring performance
DSS – Decision Support System	Middle/senior managers	Used for what-if analysis, forecasting, and projecting solutions for areas of concern; supports non-routine decisions
ESS – Executive Support System	Senior/top management	Provides text summaries and graphical dashboards comparing organisational performance to industry/competitors, for strategic decisions
KWS – Knowledge Work System	Knowledge workers (engineers, researchers)	Helps create and integrate new knowledge (e.g. CAD software)
OAS – Office Automation System	Data workers	Supports day-to-day office tasks — email, document processing, scheduling

> Memory trick — think of it as a pyramid: TPS at the bottom (operational, lots of data), MIS and DSS in the middle (management), ESS at the top (few but strategic decisions).

5.2 Characteristics of Quality Information

Characteristic	Meaning
Accurate	Free from errors, reflects reality correctly
Timely	Available when needed for decision-making
Relevant	Applies to the specific decision/problem at hand
Complete	Contains all necessary details, nothing important missing
Reliable	Can be trusted/depended on consistently
Verifiable	Can be checked/confirmed by different sources
Accessible	Easy for authorised users to obtain when needed
Secure	Protected from unauthorised access or tampering

5.3 Classical Model of Decision-Making (3 stages)

1. **Intelligence** — identifying and understanding the problem that needs a decision
2. **Design** — generating and evaluating possible alternative solutions
3. **Choice** — selecting and implementing the best alternative

Q: Explain any five types of information systems used at different levels of the organisational hierarchy.

A: TPS records daily operational transactions; MIS gives managers routine summary reports; DSS supports non-routine decisions through what-if analysis and forecasting; ESS gives top executives strategic dashboards comparing performance to competitors; KWS supports specialists in creating new knowledge (e.g. engineers using design software).

Q: Give five characteristics of quality information.

A: Accurate, timely, relevant, complete, and reliable (any 5 from the table above).

Exam Angle:

- "Explain any five/four types of information systems used at different levels" (5–10 marks)
- "Give/Describe four/five characteristics of quality information" (4–5 marks)
- "Explain any three stages of decision-making as followed in the classical model" (6 marks)
- Sometimes framed as scenario-matching: "Which type of information system would best help top management in this case, and why?"

6. Systems Development (SDLC & Waterfall Model)

6.1 Software Development Life Cycle (SDLC) — Definition

> SDLC is a structured process/framework that guides how an information system is planned, built, tested, deployed, and maintained, from the initial idea until it is retired.

6.2 Reasons an Organisation Would Develop a New Information System

- Current manual/semi-automated processes are inefficient and error-prone
- Need for real-time reporting and faster decision-making
- To improve accuracy and reduce data duplication
- To improve accessibility of records for staff/customers
- To enhance data security
- To keep up with growth (increasing volume of transactions/data)

6.3 Phases of the Waterfall Model (Winston Royce, 1970)

The Waterfall Model is **sequential** — each phase must finish before the next one begins, like water flowing downward and not going back up.

1. **Requirements Analysis** — gathering and documenting what the system must do
2. **System Design** — planning the architecture, database structure, and interface of the system
3. **Implementation (Coding)** — actual programs/software are written based on the design
4. **Testing** — checking the system for bugs/errors and confirming it meets requirements
5. **Deployment** — the system is released and put into actual use
6. **Maintenance** — ongoing fixing of bugs and updating the system after release

6.4 Disadvantages of the Waterfall Model

- Inflexible — hard to go back and change earlier phases once you've moved on
- Users only see the working system very late (at the end), so mistakes are found late
- Not ideal for large, complex projects where requirements may change along the way
- High risk if requirements were misunderstood at the start — errors cascade through all phases

6.5 Alternative Models of Software Development

Model	Description
Agile Model	Iterative, breaks the project into small cycles ("sprints"); allows continuous feedback and changes throughout development
Spiral Model	Combines iterative development with risk analysis at every cycle; good for large, high-risk projects
Prototyping Model	A working sample (prototype) of the system is built early for users to test and give feedback before the full system is built
Rapid Application Development (RAD)	Emphasises fast development using reusable components and quick prototypes, with less rigid planning
Incremental Model	The system is built and delivered in small working increments/pieces

6.6 Challenges Encountered in Systems Development

- **Lack of resources** — insufficient budget, staff, or time
- **Poor requirement gathering** — user needs not properly understood at the start
- **Resistance to change** — staff/users unwilling to adopt the new system
- **Scope creep** — uncontrolled expansion of requirements during development
- **Technical/integration challenges** — difficulty connecting the new system with existing systems
- **Poor project management/communication** among the development team

Q: Explain the different phases of the Waterfall Model of software development, as proposed by Winston Royce in 1970.

A: Requirements Analysis (gathering what the system should do) → System Design (planning the architecture/database/interface) → Implementation (writing the actual code) → Testing (finding and fixing errors) → Deployment (releasing the system for use) → Maintenance (ongoing support and fixes). Each phase must be completed before moving to the next, like water flowing downhill only.

Q: Discuss any three disadvantages of the Waterfall Model.

A: 1) It is inflexible — going back to an earlier phase to fix an error is difficult and costly. 2) Users don't see the actual working system until very late, so misunderstandings are only discovered near the end. 3) It is risky for complex projects where requirements are likely to change during development.

Exam Angle:

- "Explain the phases of the Waterfall Model" (10 marks — list and briefly explain all 6)
- "Discuss any three disadvantages of the Waterfall Model" (3 marks)
- "Describe any three alternative models of software development" (6 marks)
- "Define SDLC" (1 mark)
- "Discuss any two primary reasons an organisation would initiate development of a new system" (4 marks)
- "Explain any four phases of the systems development lifecycle" (8 marks)
- "Demonstrate any four challenges encountered in the systems development process" (8 marks)

7. Databases & SQL (the MOH Case)

7.1 Key Terms Simplified

Term	Meaning
Database	An organised collection of related data stored electronically so it can be easily accessed, managed, and updated
Table	A structure inside a database that stores data in rows (records) and columns (fields/attributes)
Primary Key	A column that uniquely identifies each row/record in a table (no duplicates, cannot be empty)
Foreign Key	A column in one table that refers to the primary key of another table — this is what creates the relationship between tables
Referential Integrity	A rule ensuring that a foreign key value must always match an existing primary key value in the related table (you cannot reference a patient that does not exist)
View	A saved, virtual "query" that displays selected data from one or more tables without storing the data itself

7.2 Worked Example — MOH (Ministry of Health) Database

Scenario: Register patients, schedule appointments, record lab tests, prescribe medication, admit/discharge patients, and bill them for services.

i. Create the database

```
CREATE DATABASE MOH;
USE MOH;
```

ii. Create at least 4 tables with appropriate data types and referential integrity

```
CREATE TABLE Patient (
    PatientID     INT PRIMARY KEY AUTO_INCREMENT,
    FirstName     VARCHAR(50) NOT NULL,
    LastName      VARCHAR(50) NOT NULL,
    DateOfBirth   DATE,
    Gender        VARCHAR(10)
);

CREATE TABLE Doctor (
    DoctorID      INT PRIMARY KEY AUTO_INCREMENT,
    DoctorName    VARCHAR(50) NOT NULL,
    Specialty     VARCHAR(50)
);

CREATE TABLE Appointment (
    AppointmentID   INT PRIMARY KEY AUTO_INCREMENT,
    PatientID       INT,
    DoctorID        INT,
    AppointmentDate DATE,
    FOREIGN KEY (PatientID) REFERENCES Patient(PatientID),
    FOREIGN KEY (DoctorID) REFERENCES Doctor(DoctorID)
);

CREATE TABLE LabTest (
    TestID        INT PRIMARY KEY AUTO_INCREMENT,
    PatientID     INT,
    TestName      VARCHAR(50),
    TestResult    VARCHAR(100),
    TestDate      DATE,
    FOREIGN KEY (PatientID) REFERENCES Patient(PatientID)
);

CREATE TABLE Billing (
    BillID        INT PRIMARY KEY AUTO_INCREMENT,
    PatientID     INT,
    TestCharges   DECIMAL(10,2),
    OperationCharges DECIMAL(10,2),
    DoctorCharges DECIMAL(10,2),
    FOREIGN KEY (PatientID) REFERENCES Patient(PatientID)
);
```

The FOREIGN KEY lines are what "enforce referential integrity" — they stop you from adding an appointment, test, or bill for a patient who doesn't exist in the Patient table.

Question 2(i): Insert at least 2 records into any four tables

```
INSERT INTO Patient (FirstName, LastName, DateOfBirth, Gender)
VALUES ('Mary', 'Namutebi', '1992-04-10', 'Female'),
       ('John', 'Okello', '1985-11-02', 'Male');

INSERT INTO Doctor (DoctorName, Specialty)
VALUES ('Dr. Kato', 'General Medicine'),
       ('Dr. Achen', 'Pediatrics');

INSERT INTO Appointment (PatientID, DoctorID, AppointmentDate)
VALUES (1, 1, '2025-08-01'),
       (2, 2, '2025-08-02');

INSERT INTO LabTest (PatientID, TestName, TestResult, TestDate)
VALUES (1, 'Malaria Test', 'Negative', '2025-08-01'),
       (2, 'Blood Sugar', 'Normal', '2025-08-02');
```

Question 2(ii): Add an extra column "Address" to the Patient table

```
ALTER TABLE Patient ADD Address VARCHAR(100);
```

Question 2(iii): Populate the new Address column

```
UPDATE Patient SET Address = 'Kampala' WHERE PatientID = 1;
UPDATE Patient SET Address = 'Mukono' WHERE PatientID = 2;
```

Question 2(iv): Change the caption for the patient's name column to "patientz_name"

> A **caption** is just the display label shown to the user — it is different from renaming the actual column. In MS Access this is done in Design View by typing a new value into the "Caption" property of the field. In plain SQL (MySQL/SQL Server), the closest equivalent is renaming the column itself:

```
-- MySQL equivalent (renaming the column, since plain SQL has no separate "caption" property)
ALTER TABLE Patient CHANGE FirstName patientz_name VARCHAR(50);

-- If using MS Access: open the Patient table in Design View, click the Name field,
-- go to the General tab, and type "patientz_name" into the Caption property box.
```

7.3 Question 3 — Creating Views

```
-- i. viewone: all details from one table
CREATE VIEW viewone AS
SELECT * FROM Patient;

-- ii. viewtwo: two columns and rows from Patient table
CREATE VIEW viewtwo AS
SELECT FirstName, LastName FROM Patient LIMIT 2;

-- iii. viewthree: number of records in a table
CREATE VIEW viewthree AS
SELECT COUNT(*) AS TotalPatients FROM Patient;

-- iv. viewfour: patients whose names start with 'M'
CREATE VIEW viewfour AS
SELECT * FROM Patient WHERE FirstName LIKE 'M%';

-- v. viewfive: patients born after 1990
CREATE VIEW viewfive AS
SELECT * FROM Patient WHERE DateOfBirth > '1990-12-31';

-- vi. viewsix: patients' names and current age
CREATE VIEW viewsix AS
SELECT FirstName, LastName,
       TIMESTAMPDIFF(YEAR, DateOfBirth, CURDATE()) AS Age
FROM Patient;

-- vii. viewseven: uses a logical operator (AND / OR / NOT)
CREATE VIEW viewseven AS
SELECT * FROM Patient WHERE Gender = 'Female' AND Address = 'Kampala';

-- viii. vieweight: multi-table query (JOIN)
CREATE VIEW vieweight AS
SELECT Patient.FirstName, Patient.LastName, Doctor.DoctorName, Appointment.AppointmentDate
FROM Patient
JOIN Appointment ON Patient.PatientID = Appointment.PatientID
JOIN Doctor ON Appointment.DoctorID = Doctor.DoctorID;
```

7.4 Traditional File Environment vs Database Management System

Problems of managing data in a traditional file environment

- **Data redundancy** — the same data is repeated/duplicated across multiple files
- **Data inconsistency** — duplicated data may be updated in one file but not another, creating conflicting values
- **Program-data dependence** — changes to data structure require changing every program that uses it
- **Lack of flexibility** — hard to generate ad-hoc reports combining data from different files
- **Poor security** — hard to control who accesses which pieces of data
- **Difficulty sharing data** across departments

How a Database Management System (DBMS) solves these problems

- Centralises data in one place, minimising redundancy and inconsistency
- Separates data from the programs that use it (data independence)
- Provides tools to enforce data integrity and validation rules
- Provides security features like user permissions and access control
- Allows flexible querying (SQL) to generate reports on demand
- Supports data sharing across the whole organisation

Exam Angle:

- Practical SQL tasks: "Create a database", "create at least 4 tables with referential integrity", "insert records", "add/rename a column", "create views" (this is the most technical, hands-on part of the course — practice typing it out, don't just read it)
- Theory: "What are the problems of managing data in a traditional file environment and how are they solved by a DBMS?" (10 marks)
- Short notes: "Write a short note on Database" (2 marks — see Section 13)

8. Networking

8.1 What is a Network?

A **network** is a group of two or more computers or devices linked together to share data, resources (like printers), and communicate with each other.

8.2 Physical (Wired) Transmission Media

Medium	Description
Twisted Pair Cable (e.g. Cat5/Cat6)	Copper wires twisted together to reduce interference; common in LANs and office networks
Coaxial Cable	A single copper conductor surrounded by insulation and shielding; used for cable TV and older networks
Fibre Optic Cable	Uses light signals through glass/plastic strands; very fast and immune to electrical interference; used for backbone/long-distance connections

8.3 Wireless Transmission Media

Medium	Description
Wi-Fi (Radio waves)	Wireless local area networking using radio frequency signals
Bluetooth	Short-range wireless connection between nearby devices
Microwave transmission	Line-of-sight signals used for long-distance wireless links between towers
Satellite	Signals relayed via satellites orbiting the earth for very long-distance/remote coverage
Infrared	Short-range wireless communication using infrared light (e.g. old TV remotes)

8.4 Network Topologies

Topology	Description
Bus Topology	All devices connect to a single central cable (the "bus"); simple and cheap but if the main cable fails the whole network goes down
Star Topology	All devices connect to a central device (switch/hub); easy to manage and one device failing doesn't affect others, but the central device is a single point of failure
Ring Topology	Devices are connected in a closed loop, data travels around the ring in one direction; if one connection breaks, the whole ring can be disrupted
Mesh Topology	Every device connects to every other device; highly reliable (many alternate paths) but expensive and complex to set up

Illustration tip: draw simple diagrams — a line with nodes hanging off it (bus), a central hub with lines to each device (star), a circle of connected devices (ring), and a web of interconnected devices (mesh).

8.5 Common Network Devices

Device	Function
Router	Connects different networks together and directs data (packets) between them, e.g. connecting a home network to the Internet
Switch	Connects multiple devices within the same network and forwards data only to the intended device
Hub	Connects multiple devices but broadcasts data to all ports (less efficient than a switch)
Modem	Converts digital signals from a computer into analog signals for transmission (and back), used to connect to the Internet via ISP lines
Network Interface Card (NIC)	Hardware component that allows a device to connect to a network
Access Point	Allows wireless devices to connect to a wired network
Firewall	Hardware/software that monitors and controls incoming/outgoing network traffic for security

Q: Explain 3 physical and 2 wireless media that you can use to connect your network.

A: Physical: twisted-pair cable (copper wires for LANs), coaxial cable (shielded copper conductor), fibre optic cable (uses light for high-speed, long-distance transmission). Wireless: Wi-Fi (radio waves for local wireless connection) and satellite (relays signals over very long distances via orbiting satellites).

Q: Describe any four network devices commonly used in organisations.

A: Router (connects and directs traffic between different networks), Switch (connects devices within one network and forwards data to the correct destination), Modem (converts digital signals to analog and back for Internet access), Firewall (monitors and filters network traffic to block unauthorised access).

Exam Angle:

- "Define a network" (1 mark)
- "Explain 3 physical and 2 wireless media" (5 marks)
- "With the aid of illustrations, explain any three different network topologies" (6 marks — draw them!)
- "Describe any four network devices commonly used in organisations" (8 marks)

9. Information Systems Security & the CIA Triad

9.1 The CIA Triad

Element	Meaning	Example of violation
Confidentiality	Only authorised people can access the information	Hackers stealing customer credit card details
Integrity	Information is accurate and has not been altered without authorisation	A hacker changing account balances in a database
Availability	Information/systems are accessible to authorised users whenever needed	A DDoS attack that takes a website offline

> Illustration: draw a triangle with "Confidentiality", "Integrity", and "Availability" at each corner, with "CIA Triad" / "Information Security" written in the middle.

9.2 Attack vs Vulnerability

Attack	Vulnerability
An actual attempt/action by someone to exploit weaknesses and cause harm to a system (e.g. a hacking attempt)	A weakness or flaw in a system that could be exploited (e.g. an outdated, unpatched password system) — it exists whether or not anyone attacks it

9.3 Common Security Weaknesses (from case studies)

- Weak/reused passwords (e.g. using "123456" across multiple systems)
- No recent data backups
- Lack of two-factor authentication
- Poor access control (too many people with unnecessary access to sensitive data)
- Outdated software with unpatched security holes
- No firewall or intrusion detection systems

9.4 Security Tools & Technologies

- **Firewalls** — filter incoming/outgoing traffic to block unauthorised access
- **Encryption** — scrambles data so it can only be read with the correct decryption key
- **Two-factor / Multi-factor Authentication (2FA/MFA)** — requires a second verification step beyond a password
- **Antivirus/Anti-malware software** — detects and removes malicious programs
- **Intrusion Detection Systems (IDS)** — monitor networks for suspicious activity
- **Regular data backups** — ensures data can be restored after loss or an attack
- **Access control lists / user permissions** — restrict who can view or edit specific data

9.5 Security Controls/Policies to Prevent Breaches

- Enforce strong, unique password policies and regular password changes
- Implement two-factor authentication for all critical systems
- Conduct regular staff security-awareness training
- Maintain regular, tested data backups
- Restrict access using the principle of least privilege (staff only access what they need)

9.6 Challenges Faced in Maintaining Information Systems Security (Digital Era)

- Constantly evolving/sophisticated cyber-threats and hacking techniques
- High cost of implementing strong security infrastructure
- Human error/negligence (e.g. clicking phishing links, weak passwords)
- Use of personal/mobile devices (BYOD) increasing entry points for attacks
- Shortage of skilled cybersecurity professionals
- Increasing interconnectivity (cloud, IoT) expanding the attack surface
- Balancing security with system usability/convenience

Q: Which elements of the CIA Triad were violated in the ABC Company case (weak reused password, data stolen, no backup)?

A: Confidentiality was violated because customer data was accessed and stolen by unauthorised hackers. Integrity could also be at risk since attackers who gain access could alter records. Availability was compromised because without a backup, the company risks permanently losing access to its data if systems are damaged.

Q: Identify two security weaknesses in a company's situation where a weak password was reused and there was no recent backup.

A: 1) Use of a weak, easily guessable password ("123456") reused across multiple systems. 2) Absence of a recent backup of the affected files, meaning data loss could be permanent.

Exam Angle:

- Case-study style questions (like ABC Company / XYZ Corporation) are very common — expect: "identify security weaknesses", "which CIA element was violated", "propose security controls", "explain challenges of securing systems"
- "With the aid of an illustration, explain the components of the CIA triad" (6 marks)
- "Demonstrate any four security tools and technologies" (8 marks)
- "Differentiate between an attack and a vulnerability" (2 marks)

10. Ethics in Information Systems

10.1 Definition

Ethics in information systems refers to the moral principles and standards that guide how individuals and organisations collect, use, store, and share data and technology responsibly.

10.2 Common Ethical Dilemmas (based on cases like GHS/MedData)

- Sharing personal data with third parties without patients'/customers' consent
- Selling personal data for profit (e.g. to pharmaceutical companies for marketing)
- Using "anonymized data" claims to justify actions while actually sharing identifiable data
- Lack of transparency — not informing clients/patients about how their data is used
- Conflicting interests between departments (Legal saying "no law was broken" vs Ethics Committee saying "trust was violated")

10.3 Ethical Principles Often Violated

Principle	Meaning
Privacy	The right of individuals to control their own personal information
Informed Consent	Individuals must knowingly agree before their data is collected/used/shared
Accountability	Organisations/individuals must take responsibility for their actions regarding data use
Transparency	Being open and honest about how data is collected and used
Property Rights	Respecting who legally "owns" data and intellectual property

10.4 Ways of Promoting Ethics in Information Systems Usage

- Develop and enforce a clear **data privacy/ethics policy**
- Obtain explicit **informed consent** from individuals before collecting/sharing their data
- Conduct regular **ethics training** for staff and IT personnel
- Set up an **ethics committee/oversight body** to review data-use decisions
- Be transparent — clearly communicate to users how their data will be used
- Implement strict access controls so only authorised staff can view sensitive data
- Establish accountability mechanisms — consequences for unethical data handling

Q: Outline any four ethical dilemmas from a scenario where a hospital shared patient data with a third party without consent.

A: 1) Sharing full identifiable patient data instead of only anonymised data as agreed. 2) Selling patient data to pharmaceutical companies for marketing without consent. 3) Failing to inform patients about the data-sharing arrangement. 4) Departments disagreeing on whether the action was acceptable, showing a lack of clear ethical accountability.

Q: As a student of information systems, advise an organisation on four ways of promoting ethics in information systems usage.

A: 1) Create and enforce a clear data-privacy policy. 2) Always obtain informed consent before collecting or sharing personal data. 3) Set up an ethics committee to oversee data-related decisions. 4) Train staff regularly on ethical data handling and hold them accountable for violations.

Exam Angle:

- Case studies are the standard format (like the GHS/MedData case) — expect: "outline ethical dilemmas", "define ethics", "explain ethical principles violated", "advise on ways to promote ethics" as a chain of questions on one scenario
- "Define ethics in the context of information systems" (2 marks)

11. E-Commerce

11.1 Definition

E-commerce is the buying and selling of goods and services, or the transmitting of data/funds, over an electronic network — mainly the Internet.

11.2 E-commerce Business Models

Model	Description	Example
B2C — Business to Consumer	Businesses sell directly to individual customers	Jumia, Amazon
B2B — Business to Business	Businesses sell/trade with other businesses	A supplier selling raw materials to a factory
C2C — Consumer to Consumer	Individuals sell directly to other individuals, usually via a platform	OLX, eBay
C2B — Consumer to Business	Individuals offer products/services to businesses	A freelancer selling design services to a company
G2C — Government to Consumer	Government provides services to citizens online	Online NIRA/URA services

11.3 Tools Used in E-commerce

Tool	Description
Internet	A global network connecting computers worldwide; the foundation for all e-commerce
Intranet	A private network internal to one organisation, used to share information among employees only
Extranet	A controlled private network that allows access to selected outsiders (e.g. suppliers, partners) as well as internal staff
Website	A collection of linked web pages where a business displays products/services and processes transactions

11.4 E-services in the Digital Era

- Search engines (Google, Bing)
- Online banking / mobile money
- Cloud storage/computing services
- Online streaming (video/music)
- E-learning platforms
- Online food/ride delivery services (e.g. SafeBoda, Jumia Food)

11.5 Challenges Hindering Effective Implementation of E-commerce

- Poor/expensive internet connectivity, especially in rural areas
- Low levels of digital literacy among some users
- Cybersecurity concerns — fear of fraud, hacking, and data theft
- Lack of trust in online payment systems
- Inadequate legal/regulatory frameworks for online transactions
- Logistics/delivery challenges, especially in areas with poor road infrastructure
- High cost of setting up e-commerce infrastructure for small businesses

Q: Explain any three e-commerce business models.

A: B2C — businesses sell directly to individual consumers (e.g. Jumia). B2B — businesses trade with other businesses (e.g. a wholesaler supplying a retailer). C2C — individual consumers sell to each other through an online platform (e.g. OLX).

Q: Outline any four challenges hindering the effective implementation of e-commerce.

A: 1) Poor internet connectivity in some areas. 2) Low trust in online payment security. 3) Limited digital literacy among users. 4) Weak legal frameworks protecting online buyers and sellers.

Exam Angle:

- "State any four models of e-commerce" (4 marks)
- "Describe the following tools as used in e-commerce: Internet, Intranet, Extranet, Website" (2 marks each)
- "Outline four examples of e-services in the digital era" (4 marks)
- "State/Outline any four challenges facing e-commerce" (4 marks)

12. Globalization & the Digital Divide

12.1 Globalization vs Digital Divide

Globalization	Digital Divide
The process by which businesses, economies, and cultures around the world become increasingly interconnected through trade, technology, and communication	The gap between people/regions who have access to modern information technology (like the Internet/computers) and those who do not

12.2 Three Stages/Eras of Globalization (Thomas Friedman)

Era	Period	Driving force
Globalization 1.0	~1492–1800	Countries globalizing — driven by countries' military/political power, exploration, and trade (e.g. colonization)
Globalization 2.0	~1800–2000	Companies/multinational corporations globalizing — driven by falling transportation and telecommunication costs (steamships, railroads, telephones)
Globalization 3.0	2000–present	Individuals globalizing — driven by the Internet and digital technology, allowing any individual anywhere to compete and connect globally

12.3 Advantages of Globalization

- Access to wider/global markets for businesses
- Faster spread of technology and innovation across countries
- Increased foreign investment and job creation
- Access to cheaper goods and a wider variety of products
- Improved cross-cultural communication and collaboration

12.4 Roles of Information Systems in Promoting Globalization

- Enable instant global communication (email, video conferencing)
- Support global supply chain management and coordination
- Allow businesses to reach global customers through e-commerce
- Enable outsourcing of work to different countries

12.5 Three Levels/Stages of the Digital Divide

Level	Meaning
Access	Whether a person has physical access to technology/internet at all (the basic gap between "connected" and "not connected")
Usage	Even with access, differences in how effectively/frequently people actually use the technology (skills gap)
Outcome/Quality of use	Whether the way people use technology actually improves their lives (e.g. using it for education/business vs only entertainment)

In the Ugandan context: many rural areas still lack reliable electricity and internet infrastructure (Access gap); some who have smartphones lack digital skills to use e-government or banking services fully (Usage gap); and those who do use the internet may only use it for social media rather than for economic opportunities (Outcome gap).

Q: Differentiate globalization and digital divide.

A: Globalization is the growing interconnection of economies and cultures worldwide through trade and technology, while the digital divide is the gap in access to and effective use of that technology between different groups or regions.

Q: Using examples from the Ugandan context, discuss the three levels of digital divide.

A: Access — many rural Ugandan communities lack reliable electricity and network coverage, so they cannot get online at all. Usage — some Ugandans own smartphones but lack the digital literacy to use services like mobile banking effectively. Outcome — those who do use the internet may use it mainly for entertainment rather than for economic or educational advancement, limiting the real benefit gained.

Exam Angle:

- "Differentiate globalization and digital divide" (2 marks)
- "Explain any three roles of information systems in promoting globalization" (6 marks)
- "Compare and contrast the three stages/eras of globalization as illustrated by Friedman" (6 marks)
- "Using examples from the Ugandan context, discuss the three levels of digital divide" (6 marks)

- "Explain four advantages of globalization" (8 marks)

13. Hardware, Software, Data & Operating Systems

13.1 Short Notes — Core Terms

Term	Simple explanation
Hardware	The physical, touchable parts of a computer system (e.g. keyboard, monitor, CPU, hard drive)
Software	The set of programs and instructions that tell hardware what to do (e.g. Windows, MS Word)
Database	An organised collection of related data stored so it can be easily accessed and managed
Data	Raw, unprocessed facts and figures with no meaning on their own (e.g. "20")
Information	Data that has been processed and organised to become meaningful and useful (e.g. "the patient is 20 years old")
Feedback	Output from a system that is used to evaluate and adjust the input/process to improve results
Digital World	The environment where information and communication are carried out through digital technology (computers, internet, mobile devices) rather than analog/manual means
E-commerce	Buying and selling goods/services electronically over the Internet

13.2 GUI vs CLI

GUI (Graphical User Interface)	CLI (Command Line Interface)
Lets users interact with a system using graphics/icons (windows, buttons, menus)	Lets users interact by typing text commands into a terminal
Easy to use, beginner-friendly	Difficult to use, needs training/memorised commands
Consumes more system memory	Consumes less memory
Requires both mouse and keyboard	Needs only a keyboard
Appearance/layout can be modified (themes, icons)	Appearance is largely fixed
Input can be entered anywhere on screen (clicking icons/menus)	Input can only be entered at the command prompt

13.3 Operating Systems

> An **Operating System (OS)** is system software that manages both computer hardware and software resources, and provides common services for computer programs to run.

Importance of an Operating System

- Enables software/program execution
- Facilitates user interaction with the computer
- Manages hardware resources (memory, CPU, storage)
- Provides essential services (file management, security)

Examples of Operating Systems

- Microsoft Windows
- macOS
- Linux/UNIX
- Android
- iOS
- Chrome OS

13.4 Components of Information Systems (recap)

Data, Hardware, Software, People, Process — see Section 3.1 for full details.

Q: Write short notes on hardware, software, database, data and information, and e-commerce.

A: Hardware — the physical devices of a computer system. Software — the programs that instruct hardware. Database — an organised collection of related data. Data and Information — data is raw unprocessed facts, while information is data that has been processed into a meaningful, useful form. E-commerce — buying and selling goods/services electronically over the internet.

Q: Differentiate a GUI from a CLI.

A: A GUI lets users interact with icons, windows, and menus using a mouse and keyboard, and is easy to use but uses more memory. A CLI

requires users to type exact text commands using only a keyboard, is harder to learn, but uses less memory.

Exam Angle:

- "Write short notes on the following: Hardware, Software, Database, Data and information, E-commerce" (2 marks each — very common recurring question)
- "Differentiate between GUI and CLI" (theory or table-comparison style)
- "Define an operating system and give examples" / "State the importance of an operating system"

14. Number Systems & Data Storage Conversions

14.1 Converting Decimal (Base 10) to Binary (Base 2)

Method: Keep dividing the number by 2, write down the remainder each time, and read the remainders from bottom to top.

Example: Convert 571 (base 10) to binary
571 ÷ 2 = 285 remainder 1
285 ÷ 2 = 142 remainder 1
142 ÷ 2 = 71 remainder 0
71 ÷ 2 = 35 remainder 1
35 ÷ 2 = 17 remainder 1
17 ÷ 2 = 8 remainder 1
8 ÷ 2 = 4 remainder 0
4 ÷ 2 = 2 remainder 0
2 ÷ 2 = 1 remainder 0
1 ÷ 2 = 0 remainder 1
Reading remainders bottom → top: **$571 = 1000111011_2$**

14.2 Converting Decimal (Base 10) to Hexadecimal (Base 16)

Example: Convert 571 (base 10) to hexadecimal
571 ÷ 16 = 35 remainder 11 → "B" (11 in hex is written as B)
35 ÷ 16 = 2 remainder 3
2 ÷ 16 = 0 remainder 2
Reading remainders bottom → top: **$571 = 23B_{16}$**

Hex digits: 0–9 stay the same, then 10=A, 11=B, 12=C, 13=D, 14=E, 15=F.

14.3 Data Storage Unit Conversions (used for the "Partition" question)

Standard conversion scale used in these exams (based on 1024, not 1000):
1 Byte = 8 bits | 1024 Bytes = 1 KB | 1024 KB = 1 MB | 1024 MB = 1 GB | 1024 GB = 1 TB

Conversion	Working	Answer
328,900 MB → GB	328,900 ÷ 1024	**≈ 321.19 GB**
50 MB → KB	50 × 1024	**51,200 KB**
67 GB → MB	67 × 1024	**68,608 MB**

Rule of thumb: Going from a bigger unit to a smaller unit (e.g. GB→MB), you **multiply** by 1024. Going from a smaller unit to a bigger unit (e.g. MB→GB), you **divide** by 1024.

Q: Give a base ten number 571. Find its equivalent in (a) Binary (b) Hexadecimal.
A: 571 in binary = 1000111011. 571 in hexadecimal = 23B. (See working above.)

Q: Convert 328900MB to GB, 50MB to KB, and 67GB to MB.
A: 328900MB ÷ 1024 ≈ 321.19GB. 50MB × 1024 = 51,200KB. 67GB × 1024 = 68,608MB.

Exam Angle:

- These usually appear as short calculation questions worth 2 marks each — show your working, not just the final answer, since marks are often given for the method.
- Practice converting a few random numbers yourself for speed before the exam.

15. Final Quick-Revision Checklist

15.1 One-Line Recall for Every Topic

- **Information System** = People + Hardware + Software + Data + Process, working to produce useful information
- **DIKW** = Data → Information → Knowledge → Wisdom (rising pyramid)
- **Core functions** = Input → Processing → Output → Storage (+ Feedback)
- **Porter's 3 strategies** = Cost Leadership, Differentiation, Focus
- **Porter's 5 forces** = New entrants, Suppliers, Buyers, Substitutes, Rivalry
- **IS types by level** = TPS (operational) → MIS/DSS (management) → ESS (top executives)
- **Quality information** = Accurate, Timely, Relevant, Complete, Reliable
- **Waterfall phases** = Requirements → Design → Implementation → Testing → Deployment → Maintenance
- **Alternative models** = Agile, Spiral, Prototyping, RAD, Incremental
- **Database keys** = Primary Key (unique ID) + Foreign Key (links tables) = Referential Integrity
- **Network topologies** = Bus, Star, Ring, Mesh
- **Network devices** = Router, Switch, Modem, Firewall
- **CIA Triad** = Confidentiality, Integrity, Availability
- **Ethics principles** = Privacy, Consent, Accountability, Transparency
- **E-commerce models** = B2C, B2B, C2C, C2B, G2C
- **Globalization eras (Friedman)** = 1.0 Countries → 2.0 Companies → 3.0 Individuals
- **Digital divide levels** = Access, Usage, Outcome
- **Conversions** = 1024 Bytes=1KB, 1024KB=1MB, 1024MB=1GB, 1024GB=1TB

15.2 How to Approach Case-Study Questions in the Exam

1. Read the case **twice** before answering — underline key facts (who, what went wrong, what technology is mentioned)
2. Match the case to the topic being tested (security breach → CIA triad/security; data sharing → ethics; new system → SDLC)
3. Always tie your answer back to specific details from the case, not just generic theory
4. For "explain/discuss" questions, give the term + a short explanation + an example from the case where possible
5. Number your points clearly and match the number of points to the marks (e.g. "any four" = give exactly 4 clear points)

15.3 Marks-to-Effort Guide

Marks	What examiners expect
1–2 marks	A short, precise definition or single fact — no need for long explanation
4–6 marks	2–4 well-explained points, each with a short explanation (1–2 sentences)
8–10 marks	4–5 points, each explained in 2–3 sentences, with examples/illustrations where relevant